Madhuri Jaiprakash Patil

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Work Authorization: F-1 OPT EAD (Valid for 3-year) | No Sponsorship Required | Open to Relocation Nationwide

PROFESSIONAL SUMMARY

Java Full Stack Software Engineer with 4+ years of professional experience building scalable enterprise applications using Java, Spring Boot, React and AWS. Proven expertise in designing RESTful and event-driven microservices, optimizing relational databases (PostgreSQL, MySQL) and deploying containerized workloads. Hands-on experience integrating AI/ML and LLM-powered features. Holds an M.S. in Computer and Information Science (GPA 4.0) from University of Michigan-Dearborn and is actively seeking full-time software engineering opportunities. Authorized to work in the U.S. without sponsorship.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL
Frontend: React, HTML5, CSS3, Responsive UI Design
Backend & APIs: Spring Boot, RESTful APIs, Event-Driven Microservices, Flask
Databases: PostgreSQL, MySQL, Amazon RDS, NoSQL — query optimization & indexing
Cloud & DevOps: AWS, Docker, GCP, CI/CD, GitHub Actions
Observability: Metrics, Logging, Distributed Tracing, Production Debugging
AI/ML: LLMs, NLP, RAG, FAISS, TensorFlow, PyTorch, Deep Learning
Practices: Agile/Scrum, TDD, Code Reviews, Clean Architecture, Customer-Focused Delivery

PROFESSIONAL EXPERIENCE

Research Assistant | University of Michigan-Dearborn, Dearborn, MI | March 2026 – Present

- Designed and executed experiments to evaluate causal discovery algorithms on simulated multi-agent interactions, applying data-driven analysis to identify relationships between vehicle behavior, trajectory and environmental factors.
- Developing simulation workflows using NVIDIA Isaac platform to design and validate multi-agent autonomous scenarios.
- Applied observability and data analysis techniques to simulation outputs, translating complex research findings into actionable insights.

Software Engineer | GenNext PLM Pvt. Ltd. — On-site at Mahindra & Mahindra Ltd., Pune, India | Sept 2019 – Apr 2023

- Designed and developed full-stack enterprise applications using Java, Spring Boot and React across 14+ dealership operation modules, directly translating product requirements into scalable technical solutions.
- Architected and implemented RESTful APIs and event-driven microservices, reducing manual data processing by 50% and improving system throughput.
- Containerized application services using Docker and deployed to cloud environments, supporting scalable, production-grade workloads.
- Optimized PostgreSQL queries and indexing strategies, improving overall system performance by 35%; debugged complex cross-service production issues to reduce incident resolution time by 35%.
- Built responsive React UI components with HTML5/CSS3/JavaScript, improving usability across dealership workflows.
- Implemented CI/CD pipelines using GitHub Actions, cutting deployment time by 40% and improving release reliability.
- Developed real-time dashboards and analytics systems, improving business decision-making speed by 40%.
- Integrated third-party REST APIs (e.g., Google Translator) to extend system capabilities and user accessibility.
- Collaborated cross-functionally in Agile/Scrum teams, improving sprint predictability and on-time delivery.

Trainee Engineer | Devise Electronics Pvt. Ltd., Pune, India |Jan 2019 – Jun 2019

- Developed embedded firmware using C, STM32, Arduino, and Raspberry Pi, improving hardware response efficiency by 20%. Conducted unit-level testing and debugging, reducing firmware defects during integration by 30%.

EDUCATION

M.S. in Computer and Information Science |GPA: 4.0
University of Michigan-Dearborn, Dearborn, MI | Apr 2023 – Mar 2026
B.E. in Electronics and Telecommunication Engineering |GPA: 3.2
Savitribai Phule Pune University, Pune, India | Aug 2013 – May 2017

PROJECTS

Resume Navigator – AI-Driven Resume Analysis | Python, LLMs, NLP, RAG, FAISS, Flask, HTML/CSS/JS

- Built an AI-powered resume screening system leveraging LLMs, embedding generation, and semantic similarity search (RAG + FAISS), reducing manual shortlisting effort by 60% and improving candidate-job matching accuracy by 50%.
- Designed Flask REST API backend with NLP preprocessing pipeline; deployed as a containerized web application.

Focus Finds – E-Commerce Platform | Java, Spring Boot, React, MySQL, REST APIs, Git

- Built a production-style full-stack e-commerce application with Spring Boot backend and React frontend, implementing authentication, catalog management, order processing, and admin workflows.
- Designed clean RESTful API interfaces following REST best practices and clean architecture principles.

CloudReads – Cloud-Based Digital Library | Java, Spring Boot, AWS, MySQL, HTML/CSS/JS

- Developed and deployed a scalable web application on AWS integrating cloud storage and RDS-backed MySQL database; architected backend services supporting multi-user access and content customization.

Event Linker – Event Networking Platform | Python, Flask, React, MySQL, AWS, AI

- Built a QR-based event networking platform with role-based access control and AI-driven attendee similarity recommendations, increasing networking engagement by 35% and reducing check-in time by 50%.

Credit Score Classification – Deep Learning | Python, TensorFlow/Keras, Pandas, Scikit-learn

- Built and evaluated an ANN-based predictive model for automated credit risk classification, including feature engineering, model training, and performance visualization.